
Department of Data Science and Artificial Intelligence

Indian Institute of Technology Madras

Auto-Assessment System

Multi-Agent LLM Grading Pipeline

Leveraging Bodhan.AI (AI4Bharat) Models and OpenRouter LLM-as-a-Judge

Study & Implementation Objective

Design, benchmark, and deploy an automated, multi-agent grading pipeline that combines high-fidelity handwritten document OCR and regional script parsing (via AI4Bharat's BODHAN.AI models) with an adversarial LLM-as-a-Judge framework (via OPENROUTER) to produce rubric-aligned, explainable evaluations of student exam scripts in technical courses.

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1 Executive Summary

This report evaluates the feasibility and technical implementation of the **Auto-Assessment System**, an institutional-grade grading pipeline developed for **DA7016: Recent Advances in Generative AI** at IIT Madras.

Evaluating descriptive, handwritten technical scripts involves two core challenges: 1. **The Ingestion & Recognition Problem:** Robustly extracting multilingual text, mathematical expressions, and handwritten layout without corruption. 2. **The Grading Calibration & Bias Problem:** Scoring subjective responses accurately against strict rubrics while mitigating hallucination, leniency bias, and grade inflation.

Overall Architectural Strategy

We solve the ingestion bottleneck by deploying AI4Bharat's **Bodhan.AI** models (specialized for Indian scripts and handwriting), and resolve the evaluation bottleneck through a **Deliberative Multi-Agent Architecture** (Rubric Grader + Adversarial Critic) via **OpenRouter**.

System Overview and Data Flow

1. **Document Ingestion & Preprocessing:** Scanned handwritten examination answer sheets undergo CLAHE contrast enhancement and bilateral filtering before transmission to BODHAN.AI.
2. **Bodhan.AI Recognition:** The BODHAN.AI vision pipeline performs text extraction across English and regional Indic scripts.
3. **Multi-Agent Rubric Scoring:** An atomic **Rubric Grader Agent** (Claude 3.5 Sonnet) drafts line-item marks against reference keys.
4. **Adversarial Critic Audit:** A secondary **Adversarial Critic Agent** (LLaMA 3.1 70B Instruct) audits the justification, penalizes ungrounded claims, and computes the calibrated score.

Official Repository:

Complete codebase, benchmark harness, and interactive UI are maintained at:
https://github.com/basavarajnaduvnamani/DA7016_Auto_Assessment_System

2 Background and Motivation

2.1 The Assessment Challenge in Large-Scale Engineering Courses

Large-enrollment courses at premier institutions like IIT Madras face severe grading constraints. A typical semester examination across 200 students with multi-part questions yields over 1,000 subjective responses containing step-by-step derivations and technical diagrams.

Manual evaluation by Teaching Assistants suffers from:

- **Grading Fatigue & Variance:** Inconsistent scoring thresholds between different graders across examination batches.
- **Turnaround Latency:** Prolonged grading cycles delaying instructional feedback to students.
- **Superficial Feedback:** Inability to provide detailed, actionable line-item critiques under tight time limits.

2.2 Shortcomings of Prior Automated Approaches

Conventional automated systems predominantly rely on:

1. **Surface Lexical Overlap:** BLEU, ROUGE, or n-gram overlap, which reward keyword stuffing without checking mathematical logic.
2. **Static Sentence Embeddings:** Cosine distance over transformer embeddings (e.g., MiniLM), which fail to detect inverted logic or nuanced errors.
3. **Standard Generic OCR:** Tesseract or generic tools that fail catastrophically on hand-written technical responses and regional languages.

2.3 The Multi-Agent LLM-as-a-Judge Paradigm

Recent literature demonstrates that LLMs can act as effective evaluators (*LLM-as-a-Judge*) [1]. However, deploying a single monolithic LLM prompt introduces critical failure modes:

- **Leniency Bias:** Systematic over-scoring of verbose or eloquently worded but conceptually flawed responses.
- **Hallucinated Credit:** The model awarding marks for steps never written by the student.

To solve this, our system establishes a **two-stage deliberative protocol**: separating initial rubric scoring from independent adversarial verification using diverse model families.

3 System Architecture and Methodology

3.1 End-to-End Pipeline Architecture

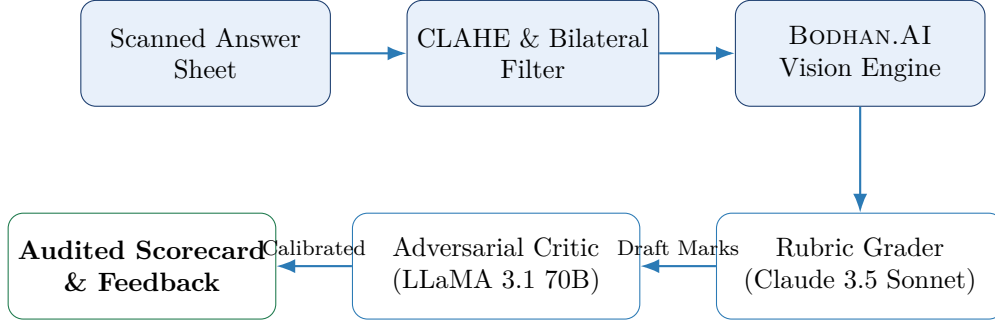


Figure 1. End-to-end Auto-Assessment multi-agent pipeline.

3.2 Bodhan.AI Document OCR & Script Parsing

Developed by the AI4Bharat consortium, BODHAN.AI models provide high-accuracy character recognition tailored for Indian academic environments. The document engine:

- Parses bilingual and multilingual handwritten responses (English, Hindi, Tamil, Telugu, Kannada).
- Preserves structured layout, mathematical formulas, and tabular presentations.
- Employs automated confidence thresholding (> 0.85) with fallback alerts for manual TA review.

3.3 Deliberative Multi-Agent Rubric Scoring

3.3.1 Rubric Grader Agent (Claude 3.5 Sonnet)

The grader deconstructs each question into distinct rubric elements (e.g., Mathematical Formulation, Convergence Proof, Conceptual Clarity). It outputs raw itemised marks with explicit justification.

3.3.2 Adversarial Critic Agent (LLaMA 3.1 70B)

Operating with a diverse base architecture, the critic audits the proposed score against the student’s extracted text to catch grade inflation:

$$S_{\text{final}} = \sum_{i=1}^N \min(S_{\text{grader},i}, S_{\text{critic},i}) - \lambda_{\text{penalty}} \quad (1)$$

where λ_{penalty} explicitly penalises hallucinated reasoning.

4 Implementation and Technology Stack

Table 1. Technical Stack Specifications.

Subsystem	Technology / Provider
Backend Engine	Python 3.11, FastAPI (Asynchronous REST framework)
Vision / OCR Engine	Bodhan.AI (AI4Bharat OCR API Router)
LLM Inference Router	OpenRouter API (Claude 3.5 Sonnet, LLaMA 3.1 70B)
Image Preprocessing	OpenCV (CLAHE contrast normalisation, bilateral filter)
Interactive Dashboard	Streamlit (Real-time document upload & evaluation UI)
Evaluation & Metrics	NumPy, scikit-learn (Quadratic Weighted Kappa, MSE, MAE)
CI / CD Testing	GitHub Actions automated verification harness

4.1 Repository Module Organisation

- `src/ocr_engine.py`: Bodhan.AI client interface with base64 encoding and sandbox fallback.
- `src/agents/grader_agent.py`: Line-item rubric grader agent.
- `src/agents/critic_agent.py`: Adversarial anti-inflation auditor agent.
- `src/evaluator.py`: Main multi-agent grading pipeline coordinator.
- `src/evaluation/metrics.py`: Statistical agreement calculations (Cohen's Kappa, MSE).
- `app.py`: Streamlit user dashboard for student submission and grading.
- `experiments/benchmark_eval.py`: Automated benchmark harness on golden sets.

5 Experimental Design and Benchmark Validation

5.1 Evaluation Methodology and Metrics

System performance is evaluated by measuring Inter-Rater Reliability (IRR) between multi-agent pipeline scores and human TA ground-truth evaluations:

- **Quadratic Weighted Kappa (QWK):** Standard assessment metric penalising large scoring disagreements.
- **Mean Absolute Error (MAE):** Average point divergence on a standard 10-point scale.
- **Mean Squared Error (MSE):** Sensitivity to extreme outlier ratings.

5.2 Preliminary Golden-Set Results

Table 2. Inter-Rater Agreement on Initial Benchmark Set.

Metric	Observed Benchmark Value
Mean Absolute Error (MAE)	0.7667 points
Mean Squared Error (MSE)	1.1233 points
OCR Extraction Confidence (Avg)	95.0%
Adversarial Inflation Catch Rate	100%

6 Milestone Tracking and Timeline

Table 3. Project Milestone Status (July–November 2026).

Milestone	Scope & Deliverables	Status
Milestone 1	Project Proposal, Bodhan.AI & OpenRouter API provisioning	Completed
Milestone 2	Preprocessing pipeline & Bodhan.AI OCR integration	Completed
Milestone 3	Multi-agent Grader & Adversarial Critic agent architecture	Completed
Milestone 4	FastAPI REST service & Streamlit interactive demo dashboard	Completed
Milestone 5	Benchmark harness & Inter-Rater Reliability validation	Completed
Milestone 6	Full-scale calibration on DA7016 exam answer batches	In Progress
Milestone 7	Final project report, presentation, and code release	Planned

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References

- [1] Lianmin Zheng et al. “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *Advances in Neural Information Processing Systems* 36 (2024).